9\. Caching and Performance Optimization

Caching is a technique used to temporarily store frequently accessed data so that subsequent requests can be served more quickly. Instead of repeatedly querying a database or calling an external API, an application can retrieve previously computed results from a cache. Redis and Memcached are commonly used caching technologies. Cache-aside is a popular caching strategy in which the application first checks the cache, retrieves data from the primary storage if the cache misses, and then stores the result in the cache. Time-to-live (TTL) settings determine how long cached entries remain valid. One significant challenge in caching is cache invalidation, which involves ensuring that outdated values are removed or updated when the underlying data changes. Poor cache invalidation strategies can cause applications to return stale or inconsistent information.

10\. Monitoring and Observability

Observability refers to the ability to understand the internal state of a system by examining its external outputs. Three commonly used observability signals are logs, metrics, and traces. Logs provide detailed records of events that occur within an application. Metrics represent numerical measurements such as request counts, CPU utilization, memory consumption, and response latency. Distributed traces show how individual requests travel through multiple services, making it easier to identify performance bottlenecks. Tools such as Prometheus, Grafana, OpenTelemetry, and Jaeger are frequently used to implement observability infrastructure. Effective monitoring systems define measurable service-level objectives (SLOs), generate alerts when important thresholds are exceeded, and provide enough context for engineers to investigate production incidents.